

Chapter 1.5 Properties of the Integral

Rick Durrett Probability: Theory and Examples

Hu Jingwu

Built from the local Chapter 1 LLMviki flash cards.

Roadmap for 1.5

Learning goal

Use Jensen, Holder, Minkowski, Fatou, MCT, DCT, and bounded convergence with exact hypotheses.

1. Inequalities: Jensen, Holder, Minkowski.
2. Limit theorems: bounded convergence, Fatou, MCT, DCT.
3. Exercise patterns: endpoint Holder, $L^p \rightarrow L^\infty$, absolute continuity, dense simple functions, series exchange.

Jensen's Inequality

Theorem

If μ is a probability measure, φ is convex, and the integrals are defined, then

$$\varphi\left(\int f d\mu\right) \leq \int \varphi(f) d\mu.$$

Let $m = \int f d\mu$. For a convex φ , choose a supporting line at m :

$$\varphi(x) \geq \varphi(m) + a(x - m).$$

Substitute $x = f$ and integrate. Since $\int (f - m) d\mu = 0$, Jensen follows.

[LLMwiki: Jensen's inequality for integrals](#); ID: durrett_ch1_jensen_integral

Holder's Inequality

Theorem

If $1 < p, q < \infty$ and $p^{-1} + q^{-1} = 1$, then

$$\int |fg| d\mu \leq \|f\|_p \|g\|_q.$$

If either norm is zero, the result is immediate. Otherwise normalize $F = |f|/\|f\|_p$ and $G = |g|/\|g\|_q$. Young's inequality gives

$$FG \leq \frac{F^p}{p} + \frac{G^q}{q}.$$

Integrating gives $\int FG d\mu \leq 1$, hence Holder.

[LLMwiki: Holder's inequality](#); ID: durrett_ch1_holder_integral

Minkowski's Inequality

Theorem

For $1 \leq p \leq \infty$,

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

For $1 < p < \infty$, write

$$|f + g|^p \leq |f| |f + g|^{p-1} + |g| |f + g|^{p-1}.$$

Apply Holder to each term and divide by $\|f + g\|_p^{p-1}$. The endpoints $p = 1$ and $p = \infty$ follow from the ordinary triangle inequality.

LLMwiki: Minkowski's inequality; ID: durrett_ch1_minkowski

Bounded Convergence

Theorem

On a finite-measure space, if $f_n \rightarrow f$ pointwise and $|f_n| \leq M$, then

$$\int f_n d\mu \rightarrow \int f d\mu.$$

Since $\mu(\Omega) < \infty$, the constant function M is integrable. Also $|f_n| \leq M$ and $|f| \leq M$ pointwise, so $|f_n - f| \leq 2M$. Apply dominated convergence.

LLMwiki: [Bounded convergence theorem](#); ID: [durrett_ch1_bounded_convergence](#)

Fatou's Lemma

Theorem

If $f_n \geq 0$, then

$$\int \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu.$$

Define $g_n = \inf_{k \geq n} f_k$. Then $g_n \uparrow \liminf_k f_k$. By MCT,

$$\int g_n d\mu \uparrow \int \liminf_k f_k d\mu.$$

Since $g_n \leq f_k$ for all $k \geq n$, $\int g_n d\mu \leq \inf_{k \geq n} \int f_k d\mu$. Let $n \rightarrow \infty$.

[LLMwiki: Fatou's lemma; ID: durrett_ch1_fatou_integral](#)

Monotone Convergence Theorem

Theorem

If $0 \leq f_n \uparrow f$, then

$$\int f_n d\mu \uparrow \int f d\mu.$$

Monotonicity gives $\int f_n d\mu$ increasing and bounded above by $\int f d\mu$. Conversely, every simple $\phi \leq f$ is eventually captured up to a small error by the increasing sequence; taking suprema over lower simple approximations gives equality.

LLMwiki: [Monotone convergence theorem](#); ID: `durrett_ch1_monotone_convergence`

Dominated Convergence Theorem

Theorem

If $f_n \rightarrow f$ a.e. and $|f_n| \leq g$ for one integrable g , then

$$\int f_n d\mu \rightarrow \int f d\mu.$$

Since $|f| \leq g$ a.e., the functions $g + f_n$ and $g - f_n$ are nonnegative. Apply Fatou to both:

$$\int (g + f) \leq \liminf \int (g + f_n), \quad \int (g - f) \leq \liminf \int (g - f_n).$$

Rearranging gives matching limsup and liminf bounds.

[LLMwiki: Dominated convergence theorem](#); ID: durrett_ch1_dominated_convergence

Exercise 1.5.1

Let

$$\|f\|_{\infty} = \inf\{M : \mu(\{x : |f(x)| > M\}) = 0\}.$$

Prove that

$$\int |fg| d\mu \leq \|f\|_1 \|g\|_{\infty}.$$

LLMwiki: Essential supremum bounds integrals; ID: exercise_method_essential_supremum_bound

Exercise 1.5.1: Proof Skeleton

Let $M = \|g\|_\infty$. By the definition of essential supremum,

$$|g| \leq M \quad \text{a.e.}$$

Hence $|fg| \leq M|f|$ a.e. Monotonicity and homogeneity give

$$\int |fg| d\mu \leq M \int |f| d\mu = \|g\|_\infty \|f\|_1.$$

LLMwiki: Essential supremum bounds integrals; ID: exercise_method_essential_supremum_bound

Exercise 1.5.2

Show that if μ is a probability measure, then

$$\|f\|_{\infty} = \lim_{p \rightarrow \infty} \|f\|_p.$$

LLMwiki: Lp norms converge to Linfty on probability spaces; ID: exercise_method_lp_to_l_infty_limit

Exercise 1.5.2: Proof Skeleton

Let $M = \|f\|_\infty$. If $M < \infty$, then $|f| \leq M$ a.e. and $\mu(\Omega) = 1$, so

$$\|f\|_p \leq M.$$

For $\varepsilon > 0$, set

$$A_\varepsilon = \{|f| > M - \varepsilon\}.$$

By essential supremum, $\mu(A_\varepsilon) > 0$. Therefore

$$\|f\|_p^p \geq (M - \varepsilon)^p \mu(A_\varepsilon), \quad \|f\|_p \geq (M - \varepsilon) \mu(A_\varepsilon)^{1/p}.$$

Let $p \rightarrow \infty$ and then $\varepsilon \downarrow 0$. If $M = \infty$, repeat with K in place of $M - \varepsilon$ and let $K \rightarrow \infty$.

LLMwiki: Lp norms converge to Linfty on probability spaces; ID: `exercise_method_lp_to_l_infty_limit`

Exercise 1.5.3

Prove Minkowski's inequality. For $p \in (1, \infty)$, use Holder's inequality to show

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Then show the same result for $p = 1$ and $p = \infty$.

LLMwiki: Minkowski's inequality; ID: durrett_ch1_minkowski

Exercise 1.5.3: Proof Skeleton

Let q be conjugate to p . If $\|f + g\|_p = 0$, stop. Otherwise,

$$\|f + g\|_p^p \leq \int |f| |f + g|^{p-1} d\mu + \int |g| |f + g|^{p-1} d\mu.$$

Holder gives

$$\int |f| |f + g|^{p-1} d\mu \leq \|f\|_p \|f + g\|_p^{p-1},$$

and the same for g . Divide by $\|f + g\|_p^{p-1}$. For $p = 1$, integrate $|f + g| \leq |f| + |g|$. For $p = \infty$, use the same pointwise inequality a.e.

[LLMwiki: Minkowski's inequality](#); ID: `durrett_ch1_minkowski`

Exercise 1.5.4

If f is integrable and E_m are disjoint sets with union E , show that

$$\sum_{m=0}^{\infty} \int_{E_m} f \, d\mu = \int_E f \, d\mu.$$

Conclude that if $f \geq 0$, then

$$\nu(E) = \int_E f \, d\mu$$

defines a measure.

LLMwiki: Monotone convergence theorem; ID: durrett_ch1_monotone_convergence

Exercise 1.5.4: Proof Skeleton

First suppose $f \geq 0$. Since E_m are disjoint,

$$\mathbf{1}_E f = \sum_{m=0}^{\infty} \mathbf{1}_{E_m} f.$$

The partial sums increase to $\mathbf{1}_E f$, so MCT gives

$$\int_E f \, d\mu = \lim_{n \rightarrow \infty} \sum_{m=0}^n \int_{E_m} f \, d\mu.$$

For signed integrable f , apply the nonnegative case to f^+ and f^- and subtract. If $f \geq 0$, the identity gives countable additivity of $\nu(E) = \int_E f \, d\mu$, with $\nu(\emptyset) = 0$ and $\nu(E) \geq 0$.

LLMwiki: [Monotone convergence theorem](#); ID: [durrett_ch1_monotone_convergence](#)

Exercise 1.5.5

If $g_n \uparrow g$ and

$$\int g_1^- d\mu < \infty,$$

show that

$$\int g_n d\mu \uparrow \int g d\mu.$$

LLMwiki: Monotone convergence theorem; ID: durrett_ch1_monotone_convergence

Exercise 1.5.5: Proof Skeleton

Set

$$h_n = g_n - g_1, \quad h = g - g_1.$$

Then $h_n \geq 0$ and $h_n \uparrow h$. By MCT,

$$\int h_n d\mu \uparrow \int h d\mu.$$

Since $\int g_1^- d\mu < \infty$, the extended integrals are well-defined and

$$\int g_n d\mu = \int g_1 d\mu + \int h_n d\mu \uparrow \int g_1 d\mu + \int h d\mu = \int g d\mu.$$

LLMwiki: [Monotone convergence theorem](#); ID: [durrett_ch1_monotone_convergence](#)

Exercise 1.5.6

If $g_m \geq 0$, show that

$$\int \sum_{m=0}^{\infty} g_m d\mu = \sum_{m=0}^{\infty} \int g_m d\mu.$$

LLMwiki: Monotone convergence theorem; ID: durrett_ch1_monotone_convergence

Exercise 1.5.6: Proof Skeleton

Let

$$s_n = \sum_{m=0}^n g_m.$$

Then $s_n \geq 0$ and $s_n \uparrow \sum_{m=0}^{\infty} g_m$. MCT gives

$$\int \sum_{m=0}^{\infty} g_m d\mu = \lim_{n \rightarrow \infty} \int s_n d\mu.$$

Finite linearity gives $\int s_n d\mu = \sum_{m=0}^n \int g_m d\mu$; let $n \rightarrow \infty$.

[LLMwiki: Monotone convergence theorem](#); ID: durrett_ch1_monotone_convergence

Exercise 1.5.7

Let $f \geq 0$.

1. Show that

$$\int f \wedge n d\mu \uparrow \int f d\mu.$$

2. Use this to conclude that if g is integrable and $\varepsilon > 0$, then there is $\delta > 0$ such that $\mu(A) < \delta$ implies

$$\int_A |g| d\mu < \varepsilon.$$

LLMwiki: Monotone convergence theorem; ID: durrett_ch1_monotone_convergence

Exercise 1.5.7: Proof Skeleton

Since $f \wedge n \uparrow f$, MCT gives

$$\int f \wedge n \, d\mu \uparrow \int f \, d\mu.$$

Apply this to $f = |g|$. Since g is integrable,

$$\int (|g| - |g| \wedge n) \, d\mu \downarrow 0.$$

Choose n so the tail is $< \varepsilon/2$. Then

$$\int_A |g| \, d\mu \leq \int_A (|g| \wedge n) \, d\mu + \int (|g| - |g| \wedge n) \, d\mu \leq n\mu(A) + \varepsilon/2.$$

Take $\delta = \varepsilon/(2n)$, with any positive δ if $n = 0$.

LLMwiki: [Monotone convergence theorem](#); ID: [durrett_ch1_monotone_convergence](#)

Exercise 1.5.8

Show that if f is integrable on $[a, b]$ and

$$g(x) = \int_{[a,x]} f(y) \, dy,$$

then g is continuous on (a, b) .

LLMwiki: Absolute continuity of the integral; ID: durrett_ch1_dominated_convergence

Exercise 1.5.8: Proof Skeleton

Fix $x \in (a, b)$ and $\varepsilon > 0$. By Exercise 1.5.7, choose $\delta > 0$ such that $\lambda(A) < \delta$ implies

$$\int_A |f(y)| dy < \varepsilon.$$

If $|h| < \delta$ and $x + h \in (a, b)$, then

$$g(x+h) - g(x) = \begin{cases} \int_{(x, x+h]} f(y) dy, & h > 0, \\ -\int_{(x+h, x]} f(y) dy, & h < 0. \end{cases}$$

In both cases the absolute value is bounded by the integral of $|f|$ over an interval of length $|h| < \delta$, hence is $< \varepsilon$.

LLMwiki: Absolute continuity of the integral; ID: durrett_ch1_dominated_convergence

Exercise 1.5.9

Show that if

$$\|f\|_p = \left(\int |f|^p d\mu \right)^{1/p} < \infty,$$

then there are simple functions ϕ_n such that

$$\|\phi_n - f\|_p \rightarrow 0.$$

LLMwiki: Simple functions are dense in L_p ; ID: `exercise_method_density_of_simple_functions_in_lp`

Exercise 1.5.9: Proof Skeleton

For $1 \leq p < \infty$, choose simple ϕ_n such that

$$\phi_n(x) \rightarrow f(x), \quad |\phi_n(x)| \leq |f(x)|.$$

For example, approximate f^+ and f^- by dyadic simple functions from below. Then

$$|\phi_n - f|^p \leq (|\phi_n| + |f|)^p \leq 2^p |f|^p.$$

Since $\|f\|_p < \infty$, the dominating function $2^p |f|^p$ is integrable. DCT gives

$$\int |\phi_n - f|^p d\mu \rightarrow 0,$$

so $\|\phi_n - f\|_p \rightarrow 0$.

LLMwiki: Simple functions are dense in L_p ; ID: `exercise_method_density_of_simple_functions_in_lp`

Exercise 1.5.10

Show that if

$$\sum_n \int |f_n| d\mu < \infty,$$

then

$$\sum_n \int f_n d\mu = \int \sum_n f_n d\mu.$$

LLMwiki: Exchange integral and absolutely summable series; ID:

`exercise_method_absolute_summability_exchange`

Exercise 1.5.10: Proof Skeleton

By Exercise 1.5.6 applied to $|f_n|$,

$$\int \sum_n |f_n| d\mu = \sum_n \int |f_n| d\mu < \infty.$$

Thus $H = \sum_n |f_n|$ is integrable and finite a.e. Let

$$S_N = \sum_{n=0}^N f_n, \quad S = \sum_n f_n.$$

Then $S_N \rightarrow S$ a.e. and $|S_N| \leq H$. DCT gives

$$\int S_N d\mu \rightarrow \int S d\mu.$$

Since finite linearity gives $\int S_N d\mu = \sum_{n=0}^N \int f_n d\mu$, let $N \rightarrow \infty$.

LLMwiki: [Exchange integral and absolutely summable series](#); ID: `exercise_method_absolute_summability_exchange`